



Machine learning-enhanced multi-gas discrimination with a miniaturized MOS sensor array

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Abstract

Accurate and miniaturized gas sensing has become increasingly essential for real-time environmental monitoring and industrial safety. This study proposes a computationally enhanced microelectronic gas sensing platform that combines a compact metal oxide semiconductor (MOS) sensor array with machine learning algorithms for selective multi-gas detection. The core of the system is the MICS4514 sensor, which integrates two miniaturized MOS sensing elements into a single package, enabling detection of carbon monoxide, nitrogen dioxide (NO₂), ammonia (NH₃), and hydrogen (H₂) across various concentration levels. Sensor output data were processed using several supervised machine learning models, including Decision Tree, Random Forest, Quadratic Discriminant Analysis (QDA), and Gradient Boosting. While QDA yielded the highest accuracy in initial classifications, data augmentation strategies significantly improved GB's performance, achieving 100% accuracy in gas discrimination. In addition, linear regression analysis was employed to estimate gas concentrations, demonstrating its feasibility for quantitative sensing. This integration of microscale sensor technology and data-driven computational modeling underscores the potential of embedded intelligence in low-power, cost-effective gas sensors. The approach presented here supports the development of scalable on-chip sensing solutions for smart electronics and Internet-of-Things (IoT)-enabled environmental surveillance systems.

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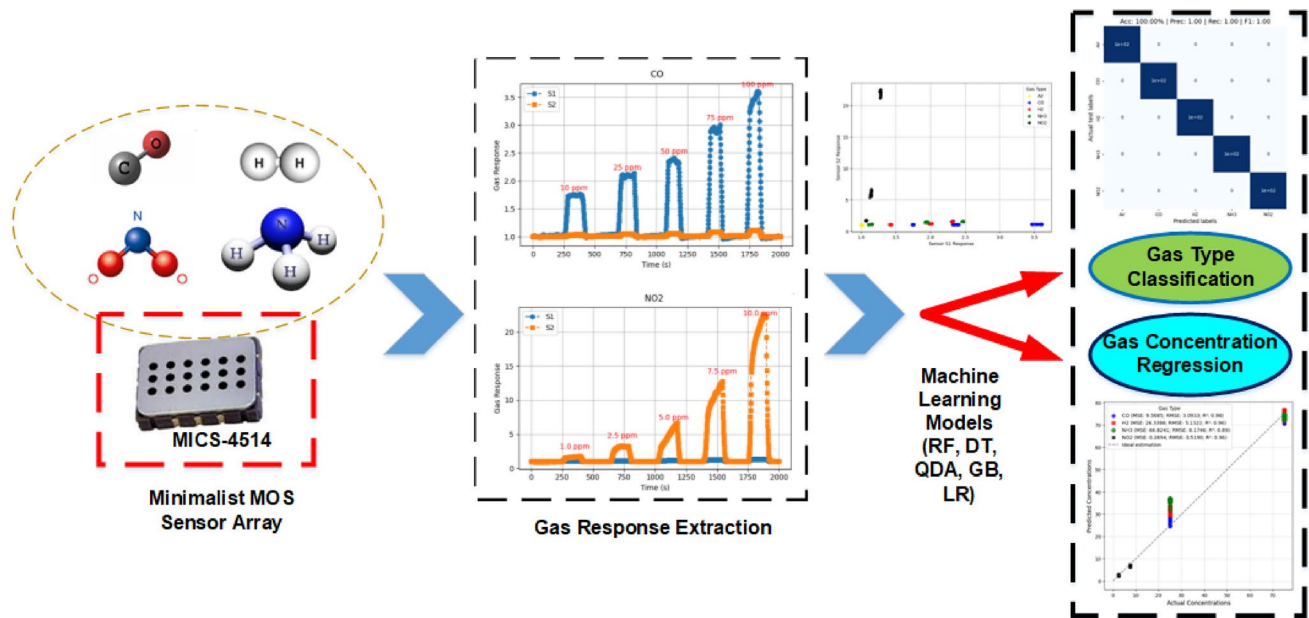
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Graphical Abstract



Keywords Air quality monitoring · Data augmentation · Gas discrimination · Machine learning · MOS sensors

1 Introduction

Air pollution is a critical global challenge with severe environmental and health impacts. [1–3]. The increase in industrial activities, transportation, and urbanization has resulted in the emission of significant amounts of pollutants into the atmosphere, adversely affecting human health and ecosystems [4]. In this context, the timely detection and identification of toxic gases in the environment are of utmost importance. Hazardous gases such as carbon monoxide (CO), nitrogen dioxide (NO₂), ammonia (NH₃), and hydrogen (H₂) pose significant health risks even at low concentrations [5–10]. Thus, developing a cost-effective and accurate air quality monitoring system is vital for public health and environmental safety.

The development of toxic gas monitoring systems often requires multiple sensors to detect and differentiate target gases [11]. However, deploying numerous sensors significantly increases costs and system complexity, particularly in mobile applications or resource-constrained environments. Among various gas sensors, Metal-Oxide Semiconductor (MOS) sensors are widely studied due to their high sensitivity, fast response time, and relatively low cost. Despite these advantages, MOS sensors also have some limitations, such as susceptibility to humidity and temperature variations, as well as cross-sensitivity to different gases [12–15]. An alternative approach involves using an

electronic nose (e-nose) [16–18], which employs a limited number of cross-sensitive sensors combined with pattern recognition techniques to enhance gas identification. The e-nose design offers several advantages, including reduced hardware costs, lower power consumption, and greater feasibility for integration into compact or portable devices.

Additionally, by leveraging advanced signal processing and machine learning algorithms, an e-nose can achieve acceptable levels of accuracy even with a small sensor array [19, 20]. However, this approach also presents certain limitations. A limited number of sensors may lead to reduced gas discrimination capability due to overlapping sensor responses, making it difficult to distinguish gases with similar chemical properties. Moreover, lower accuracy and reliability in gas identification can arise from constrained data availability, requiring sophisticated calibration and compensation techniques. Furthermore, environmental factors such as temperature, humidity, and sensor drift can further complicate the classification process, necessitating robust algorithms for real-time adaptation and error correction [21]. As a result, developing efficient methods for accurately identifying toxic gases using a minimal number of sensors remains a critical research focus, particularly in applications where cost, power efficiency, and portability are essential considerations.

In this study, we propose a method for identifying four types of toxic gases using the MICS4514 gas sensor, which integrates two high-sensitivity MOS sensors. The

MICS4514 sensor was selected for its ability to detect multiple gases at a low cost and its compact design, making it suitable for mobile environmental monitoring and resource-constrained settings. Sensor response data were collected at various concentration levels for each target gas. To enhance model robustness, data augmentation techniques [1], including Gaussian noise addition [22], baseline drift simulation [23, 24], and interpolation of intermediate concentrations, were applied exclusively to the initial training dataset. This augmentation process introduced controlled synthetic variations to better simulate real-world fluctuations in sensor readings, ensuring that the trained models remained reliable despite minor deviations in data. By validating model performance under these augmented conditions, this study demonstrates the feasibility of achieving high classification accuracy using a compact sensor array.

Several advanced machine learning models were utilized for gas classification [25], including Decision Tree (DT), Random Forest (RF), Quadratic Discriminant Analysis (QDA), and Gradient Boosting (GB) [26]. The QDA model initially achieved the highest classification accuracy when trained on the original dataset. However, after augmenting the training set, the accuracy of GB and DT significantly improved, ultimately surpassing QDA. Notably, the GB model achieved 100% classification accuracy, demonstrating that accurate toxic gas identification is feasible with only two MOS sensors when appropriate data processing techniques are applied. Additionally, a linear regression model was utilized for estimating gas concentrations [27–29]. The proposed method effectively minimizes the cost and complexity of toxic gas monitoring systems while ensuring

robust classification performance. These findings highlight the potential of integrating machine learning and data augmentation techniques for real-world applications in air quality monitoring and early toxic gas detection.

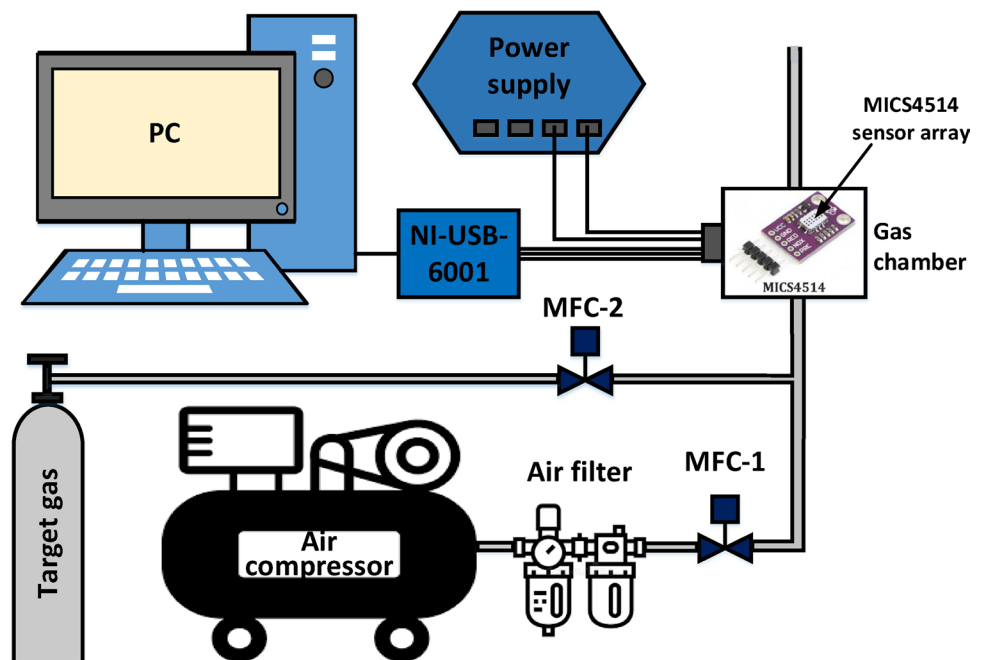
2 Materials and methods

2.1 Sensor selection and experimental setup

The MICS4514 sensor, with dual MOS elements, was chosen for its capability to detect multiple gases, including carbon monoxide (CO), nitrogen dioxide (NO₂), ammonia (NH₃), hydrogen (H₂), and several vapor organic compounds (VOCs). This sensor is compact, cost-effective, and well-suited for real-time environmental monitoring applications. The sensor chip was placed inside a controlled gas chamber to ensure stable experimental conditions and minimize external interference.

The experimental setup, as illustrated in Fig. 1, consists of several key components. A target gas cylinder supplies the desired gas, which is regulated by mass flow controllers (MFC-1 and MFC-2) to precisely control the gas concentration before entering the chamber. An air compressor, coupled with an air filter, ensures clean air is mixed with the target gas, achieving the desired gas concentrations. The output pins of the MOS sensors are connected to a NI-USB-6001 data acquisition unit, which transmits sensor measurement data to a computer for data recording and further processing. A dedicated power supply provides stable operation

Fig. 1 Block diagram of the experimental system



for all electronic components, ensuring consistent sensor performance.

The sensor response was evaluated under controlled concentrations of reducing gases (CO , NH_3 , H_2) and an oxidizing gas (NO_2). Reducing gases were tested at concentrations ranging from 10 to 100 ppm, while the oxidizing gas was introduced at lower concentrations (1–10 ppm) due to its higher reactivity. Each gas exposure was maintained for a fixed duration to allow the sensor response to stabilize. The chamber was purged with clean air between tests to reset the sensor baseline. All experiments were conducted under controlled laboratory conditions to ensure measurement repeatability and consistency. The controlled setup and systematic data collection enabled a reliable assessment of the sensor's gas discrimination capability, forming a foundation for developing robust machine-learning models for multi-gas classification.

2.2 Gas responses dataset

Data were collected systematically for reliable sensor response measurements. The MOS sensors were exposed to controlled concentrations of target gases inside the gas chamber, with response data recorded in real-time using the data acquisition system. Each measurement session included an initial baseline period in clean air, followed by gas exposure and a recovery phase. The sensor's electrical resistance values were continuously logged at a fixed sampling rate. To standardize responses, we normalized resistance ratios as R_a/R_g for reducing gases (CO , NH_3 , H_2) and R_g/R_a for the oxidizing gas (NO_2), where R_a and R_g denote resistances in clean air and target gas, respectively. This mitigated sensor drift and baseline variability. Given the disparate dynamic ranges of $S1$ (~ 1 – 3.5) and $S2$ (~ 1 – 25), normalization parameters were derived solely from the training set and applied to the test set. This is a critical step for scale-sensitive models like QDA to prevent data leakage.

In this study, four target gas types were investigated using the sensor array. Five distinct concentration levels were tested for each gas. For each concentration level, five independent and randomized exposure sessions were conducted. During each session, 10 consecutive data points were recorded, resulting in 250 data points per gas. Between gas exposures, an equal number of baseline measurements (clean air exposure) were performed to ensure complete sensor recovery and monitor potential drift. This protocol generated 250 data points per gas plus 250 baseline measurements, yielding a total dataset of 1,250 points. The dataset was then split into a training set and a test set. The training set consisted of 750 data points, including 150 points for baseline air and 150 points for each gas at three selected concentrations (lowest, highest, and mid-range). The remaining 500 data points formed the test set, comprising 100 baseline

air points and 100 points for each gas at two concentrations different from those in the training set. Three data augmentation techniques (Gaussian noise addition, baseline drift simulation, and concentration interpolation) were applied exclusively to the training set to improve model robustness. Each method generated 1,000 synthetic data points, expanding the training set from 750 original samples to 3,750 points. The test set, consisting of 500 data points, was kept entirely unmodified to ensure an unbiased evaluation. This strategy effectively prevents data leakage and allows for an accurate assessment of each model's generalization capability. Gaussian noise was added to simulate sensor fluctuations, while baseline drift was introduced to account for long-term sensor instability. Additionally, concentration interpolation was performed to generate synthetic data points between measured concentrations, ensuring a smoother representation of gas response patterns. These augmentation methods effectively doubled the size of the training dataset, leading to improved generalization of the machine learning models used for multi-gas classification.

2.3 Machine learning models for gas analysis

Four advanced machine learning models were employed to classify the sensor response data and accurately identify target gases: Decision Tree (DT) [30, 31], Random Forest (RF) [31–33], Quadratic Discriminant Analysis (QDA) [34], and Gradient Boosting (GB) [26, 35, 36]. Each model offers distinct advantages in terms of interpretability, robustness, and classification performance.

Decision Tree (DT): A hierarchical model that partitions data based on feature values, offering high interpretability but is susceptible to overfitting.

Random Forest (RF): An ensemble of multiple decision trees that improves classification accuracy and generalization by reducing overfitting.

Quadratic Discriminant Analysis (QDA): A probabilistic model that accounts for class-specific covariance structures, making it well-suited for datasets with complex distributions.

Gradient Boosting (GB): A sequential learning approach where each new model corrects the errors of previous models, leading to high accuracy and robustness.

Each model was implemented and evaluated in Python using the scikit-learn library, running in a Google Colab environment. The data were split into training and testing sets in a cross-validation manner to evaluate generalization, training on one dataset and evaluating on another to better simulate real-world deployment scenarios. The target classes included five categories: Air, CO , H_2 , NH_3 , and NO_2 . The DT classifier was used as a baseline due to its simplicity and interpretability. The RF model, an ensemble of decision trees, was implemented with a maximum depth of 2 to avoid

overfitting and improve generalization on unseen data. The QDA was selected for its effectiveness in modeling class-conditional distributions under the assumption of normality. The GB, known for its strong predictive performance, was configured with 100 estimators, a learning rate of 0.05, and a depth of 5 to balance bias and variance.

For each model, predictions were visualized using confusion matrices, and performance was reported in terms of key classification metrics [26, 37, 38]:

Accuracy (Acc): The ratio of correctly classified gas samples to the total number of samples.

Recall (Rec): The ability to correctly detect positive cases (true positive rate).

Precision (Prec): The proportion of correctly identified positive predictions.

F1-score (F1): The harmonic mean of precision and recall, providing a balanced performance measure.

Additionally, linear regression was applied to model the relationship between sensor responses and gas concentrations. The regression model's performance was evaluated using Mean Squared Error (MSE) [39, 40], Root Mean Squared Error (RMSE) [41], and the coefficient of determination (R^2) [42–44]. MSE quantifies the average squared

error, RMSE provides an interpretable error measure in the same unit as the target variable, and R^2 indicates the proportion of variance explained by the model. These evaluation metrics ensured a comprehensive assessment of both classification and regression models, providing insights into their suitability for multi-gas detection and concentration estimation.

3 Results and discussions

3.1 Gas responses and data augmentation

Figures 2 and 3 illustrate the gas response characteristics of the sensor array when exposed to four different gases: CO, H₂, NH₃, and NO₂. The response of each sensor element (S1 and S2) is represented as the normalized resistance ratio (R_a/R_g for reducing gases CO, H₂, and NH₃; R_g/R_a for the oxidizing gas NO₂) at various concentrations. The dynamic responses over time (Fig. 2) and the steady-state response values at various concentrations (Fig. 3) highlight the distinct sensing behaviors of sensor elements S1 and S2. In the case of CO, H₂, and NH₃, S1 exhibits significantly stronger

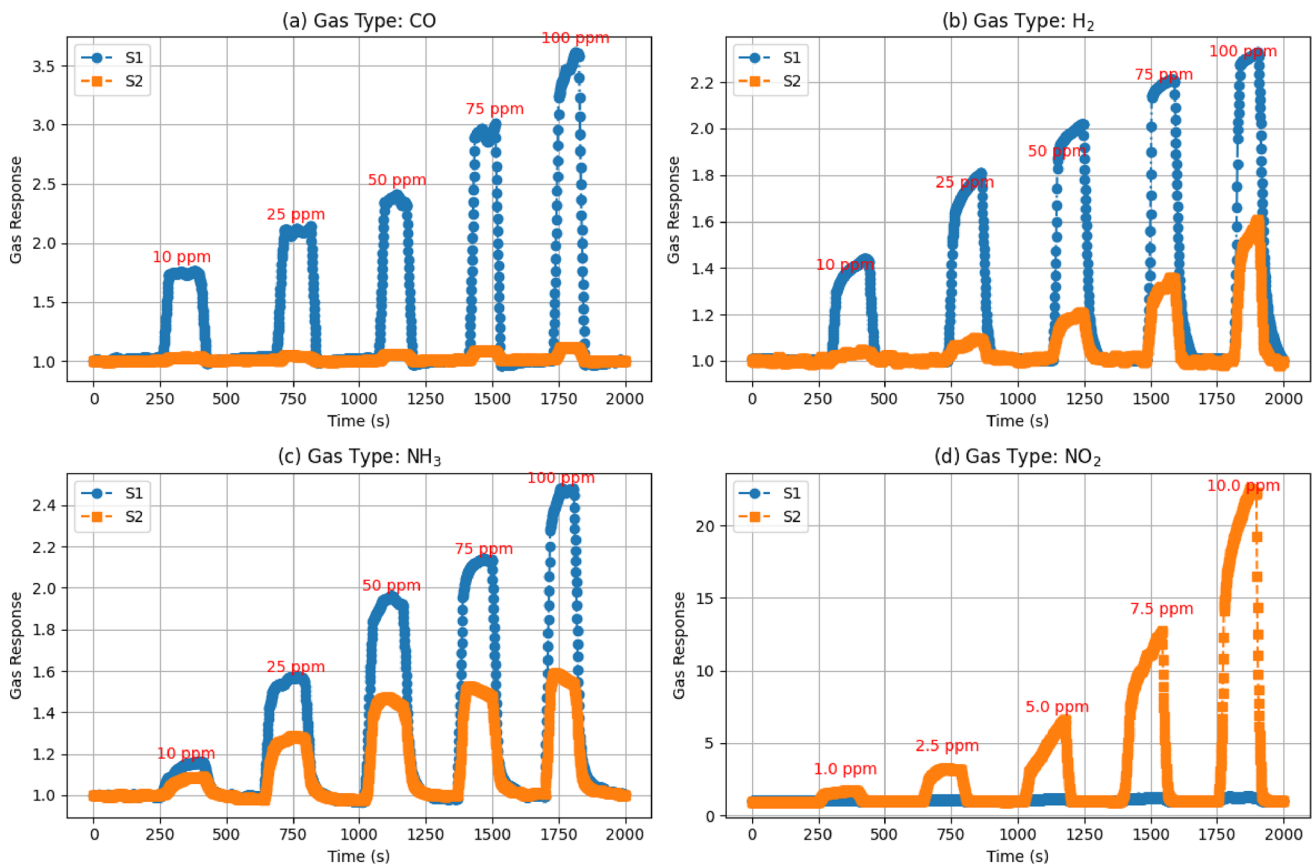


Fig. 2 Dynamic gas responses of the sensor array to various gases of **a** CO, **b** H₂, **c** NH₃, and **d** NO₂

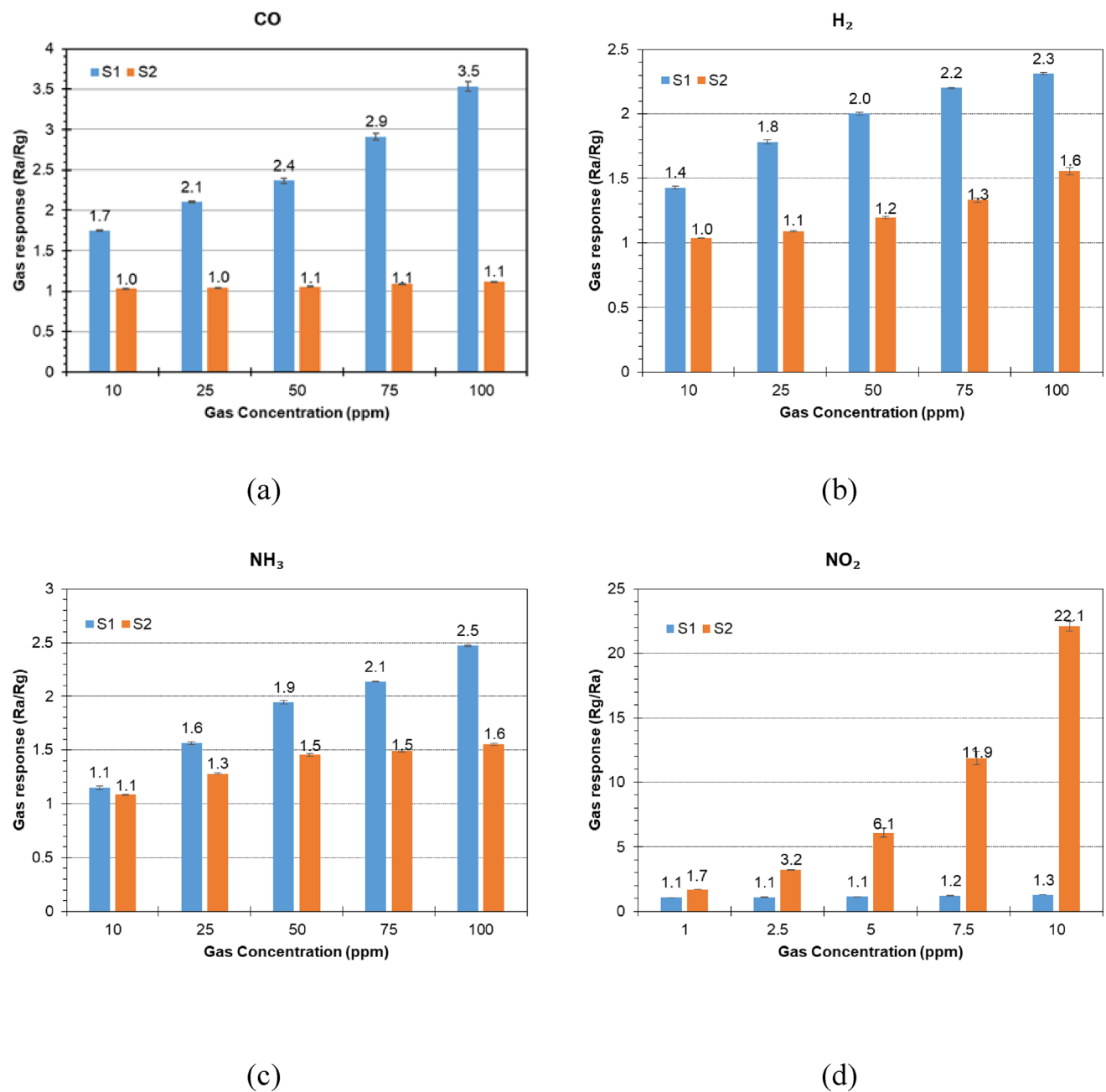


Fig. 3 Gas responses of the sensor array to various gases of **a** CO, **b** H₂, **c** NH₃, and **d** NO₂

responses than S2, with peak values increasing as the concentration rises. Specifically, for CO, S1's response grows from 1.7 at 10 ppm to 3.5 at 100 ppm, while S2 remains relatively stable around 1.1. Similarly, for H₂, S1's response reaches 2.3 at 100 ppm, whereas S2 remains much lower at 1.6. NH₃ follows the same trend, with S1 showing a stronger response (1.1 to 2.5) compared to S2 (1.1 to 1.6), confirming that S1 is the more effective element for detecting these reducing gases. In contrast, the response to NO₂ is dominated by S2, which exhibits a dramatic increase from 1.7

at 1 to 22.1 at 10 ppm, whereas S1's response remains low (1.1–1.3). This indicates that S1 is highly sensitive to reducing gases (CO, H₂, NH₃), while S2 is particularly responsive to the oxidizing gas NO₂. However, the overlapping responses of S1 to CO, H₂, and NH₃ suggest a challenge in selectivity, necessitating advanced data processing techniques for accurate gas classification.

This sensor behavior plays a crucial role in effective gas classification, especially in applications that require detecting multiple gases using a minimalist sensor array. However,

the significant overlap in responses to reducing gases such as CO, NH₃, and H₂ indicates that the sensor array still lacks high selectivity. To enhance gas selectivity and classification, data augmentation techniques can be applied. Furthermore, implementing pattern recognition models using machine learning techniques can improve classification accuracy by distinguishing response patterns across different gases. Feature engineering approaches, including statistical analysis of sensor responses, can further enhance gas differentiation. These strategies are essential for refining the sensor array's performance and ensuring robust gas detection and classification in real-world applications.

To enhance gas classification performance, data augmentation techniques were applied to the gas sensor response dataset by introducing Gaussian noise, baseline drift, and concentration interpolation to increase dataset diversity and robustness. Figure 4 presents scatter plots of Sensor S1 vs. Sensor S2 responses for five gas types (Air, CO, H₂, NH₃, and NO₂), where the original dataset (Fig. 4a) shows limited separation between gas clusters, particularly for reducing gases (CO, H₂, NH₃), due to the small sample size and inherent sensor noise. After data augmentation (Fig. 4b), the gas response distribution becomes denser and more structured, improving representation and separation between different gas types. The augmented dataset was generated by adding small Gaussian noise (mean = 0, standard deviation = 0.02) to simulate realistic sensor variations, along with baseline drift and concentration interpolation to further expand the dataset. This augmentation approach enhances gas cluster separation, balances the dataset, and reduces the risk of overfitting, enabling machine learning models to generalize better to unseen gas concentrations. Additionally, the

complementary nature of the S1 and S2 sensors, where S1 is more responsive to reducing gases and S2 is highly sensitive to the oxidizing gas NO₂, plays a crucial role in distinguishing gas types. In the next phase, both the original and augmented datasets will be used to train classification models, aiming for improved accuracy and robustness in gas identification tasks.

3.2 Gas discrimination capability using advanced machine learning models before and after data augmentation

The classification performance of four machine learning models (RF, DT, QDA, and GB) was evaluated using the original sensor response datasets. Figure 5 presents the confusion matrices, detailing the accuracy, recall, precision, and F1-score of each model. The results indicate that QDA achieves a perfect classification accuracy of 100%, successfully distinguishing all five gas types. This demonstrates the model's ability to effectively capture the unique patterns in the sensor responses, making it highly reliable for gas identification tasks. In contrast, GB and DT classifiers exhibit moderate performance, with accuracies of 70% and 68.8%, respectively, but still struggle to differentiate between NH₃ and H₂, likely due to overlapping feature distributions. RF demonstrates the lowest accuracy at 60%, with significant misclassification, particularly misidentifying all H₂ and NH₃ samples as CO, which suggests that RF fails to extract sufficiently discriminative features for these gases. These findings highlight the superior performance of QDA in handling the original sensor data, while ensemble-based models like GB and DT show promise but require further improvements.

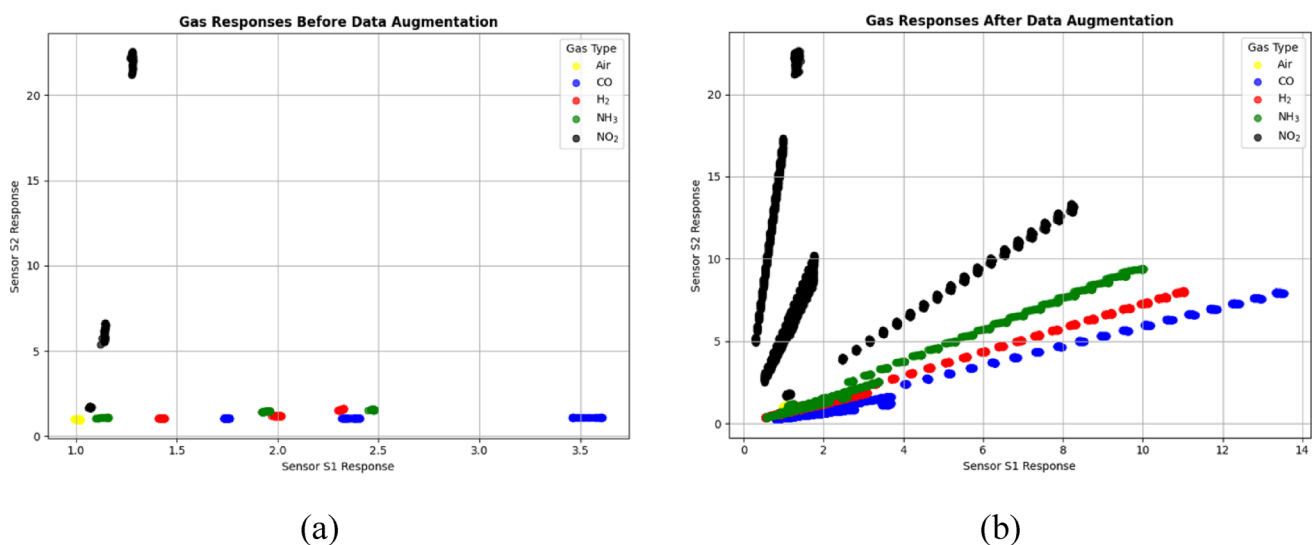


Fig. 4 Gas sensor responses before and after data augmentation using Gaussian noise, baseline drift, and concentration interpolation. Scatter plots of Sensor S1 vs. S2 responses **a** before and **b** after data augmentation

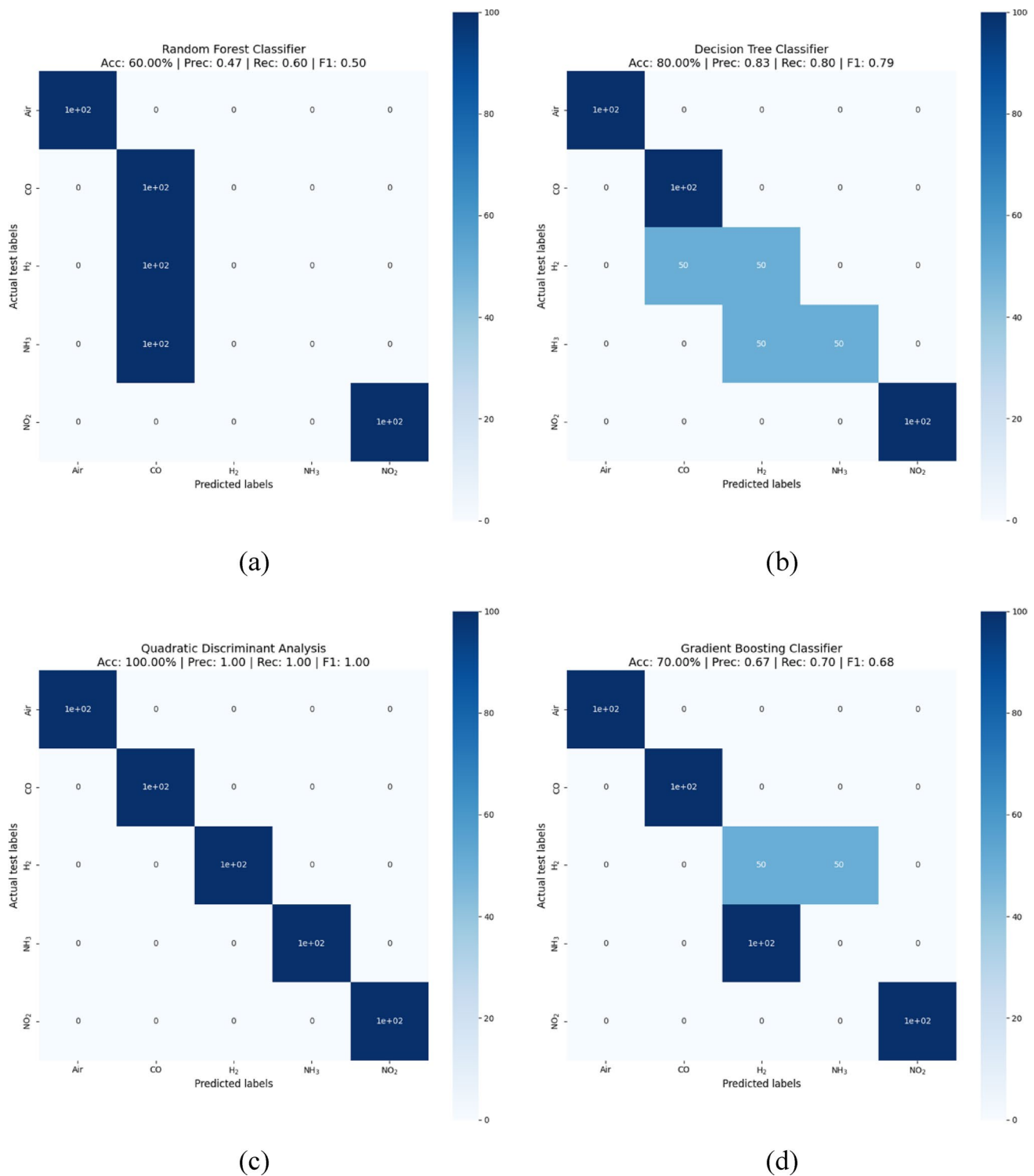


Fig. 5 Confusion matrices of various machine learning models trained and tested on the original datasets for gas classification: **a** RF classifier, **b** DT classifier, **c** QDA classifier, and **d** GB classifier

Additional techniques such as feature selection, parameter tuning, or incorporating domain knowledge could enhance their classification performance and improve their ability to distinguish between similar gases.

Figure 6 illustrates the confusion matrices of four machine learning models trained on a fully augmented dataset consisting of 3,000 synthetic samples (1,000 each from Gaussian noise addition, drift simulation, and concentration

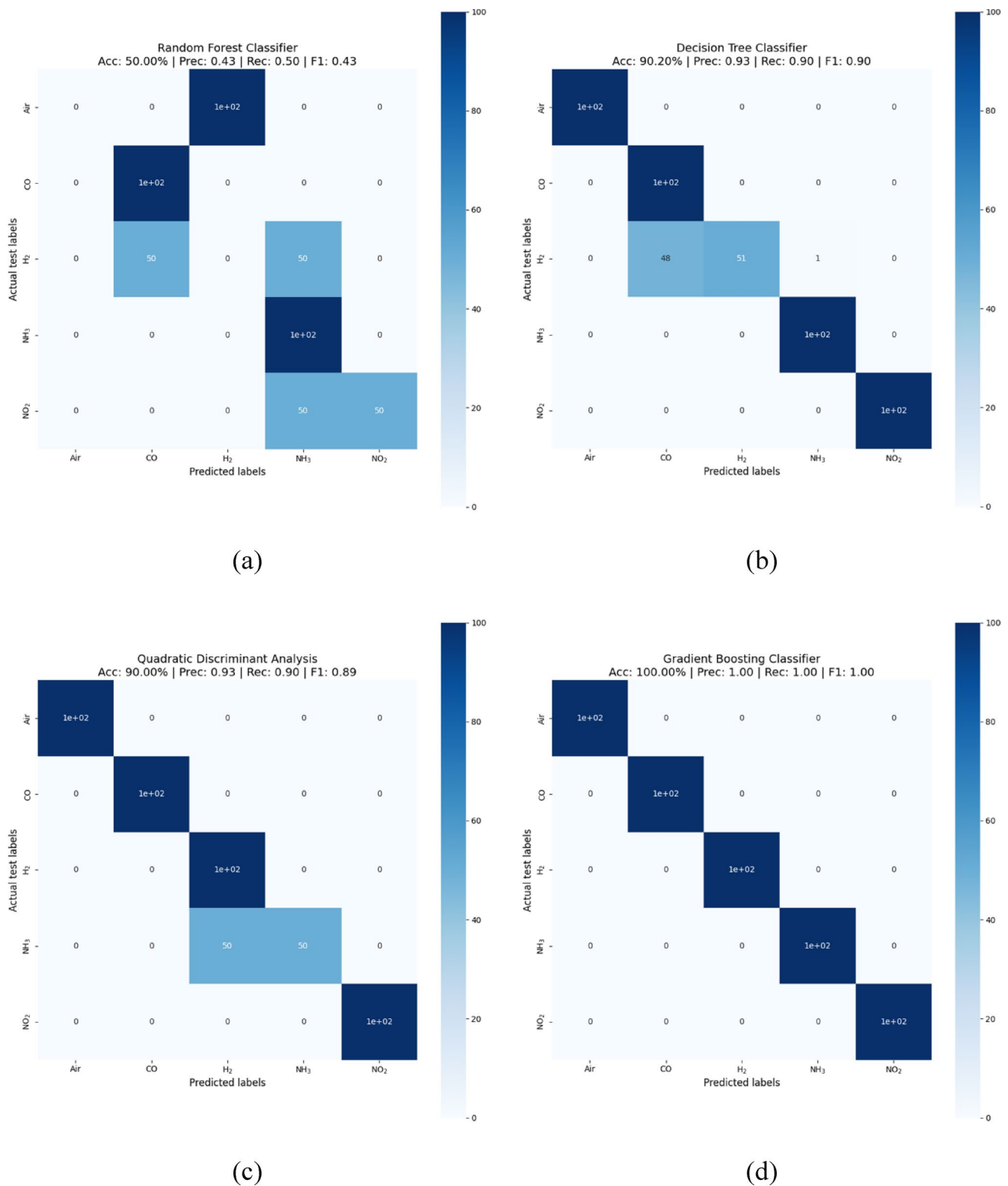


Fig. 6 Confusion matrices of various machine learning models trained on the fully augmented dataset for gas classification in the original test set: **a** RF classifier, **b** DT classifier, **c** QDA classifier, and **d** GB classifier

interpolation) combined with the original 750 training points, and evaluated using the untouched 500-sample test set to ensure unbiased assessment. The results demonstrate significant performance improvements, particularly for DT and GB models. GB achieves a perfect 100% classification accuracy after augmentation, while DT reaches 90.2% with only minor misclassifications between chemically similar gases like NH_3 and H_2 . QDA maintains strong performance at 90% accuracy, though some confusion persists among reducing gases, indicating that while augmentation improves generalization, it doesn't completely resolve feature-space overlaps for gases with similar chemical properties. In contrast, RF performance drops to 50% accuracy, likely due to overfitting on synthetic patterns, underscoring how different models respond variably to augmentation.

Figure 7 presents a detailed comparison of model performance under various training data augmentation strategies. For comparison, confusion matrices corresponding to models trained with each augmentation method are provided in Figures S1, S2, S3. The results reveal that GB demonstrates the most remarkable improvement, increasing from 70.0% accuracy with the original dataset to a perfect 100.0% classification accuracy when all augmentation methods are combined. DT shows particular sensitivity to drift augmentation, achieving 90.0% accuracy with this method alone and peaking at 90.2% with full augmentation. Interestingly, QDA, despite initially achieving perfect accuracy on the original data, experiences slight declines to 90.0–95.2% with certain augmentation techniques, suggesting that its statistical assumptions may be influenced by synthetic data distributions. RF exhibits the most variable performance, improving to 70.0% with drift augmentation but dropping significantly to 30.0–50.0% with other methods. These findings indicate

that drift simulation is the most effective individual augmentation technique across models, whereas interpolation appears less reliable, particularly for RF. The results highlight two key insights: (1) ensemble tree-based methods (GB and DT) benefit substantially from data augmentation, and (2) each model type exhibits unique sensitivity to specific augmentation approaches. This underscores the importance of developing customized augmentation strategies tailored to the characteristics of each model for optimal gas detection performance. To further validate these findings and enhance the evaluation of model generalization, K-fold cross-validation will be considered in future work [32, 45]. By partitioning the dataset into K subsets and rotating the test set across folds, this method offers a more reliable estimate of classification performance across diverse data splits, helping reduce bias and mitigate overfitting effects.

3.3 Gas concentration estimation

The scatter plot in Fig. 8 illustrates the relationship between the predicted and actual gas concentrations for four gases (CO , H_2 , NH_3 , and NO_2) using a linear regression model [28, 29] on the original train and test datasets. The dashed line represents the ideal estimation, while the colored markers show the predicted values for each gas. The performance evaluation, based on Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score, reveals significant variations in prediction accuracy across different gases.

Notably, the model achieves the highest accuracy for NO_2 , with an exceptionally low RMSE of 0.5190 and an R^2 score of 0.96, indicating minimal deviation from the true concentrations. This superior performance suggests that NO_2 exhibits a distinct sensor response, allowing the

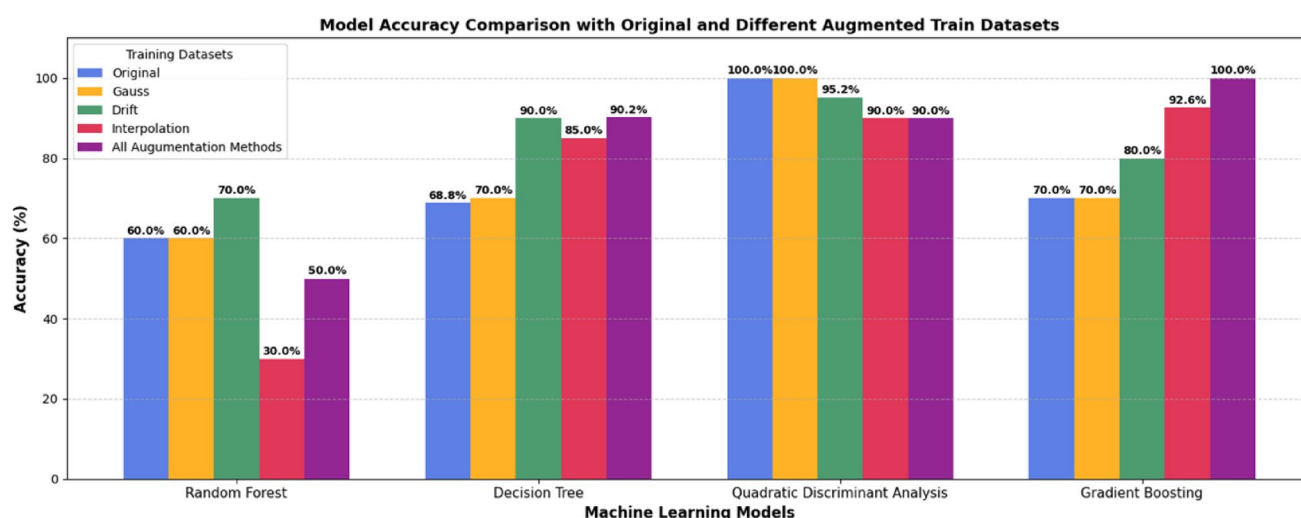


Fig. 7 Comparison graph of identification accuracy on the test dataset for machine learning models trained with the original and different augmented train datasets

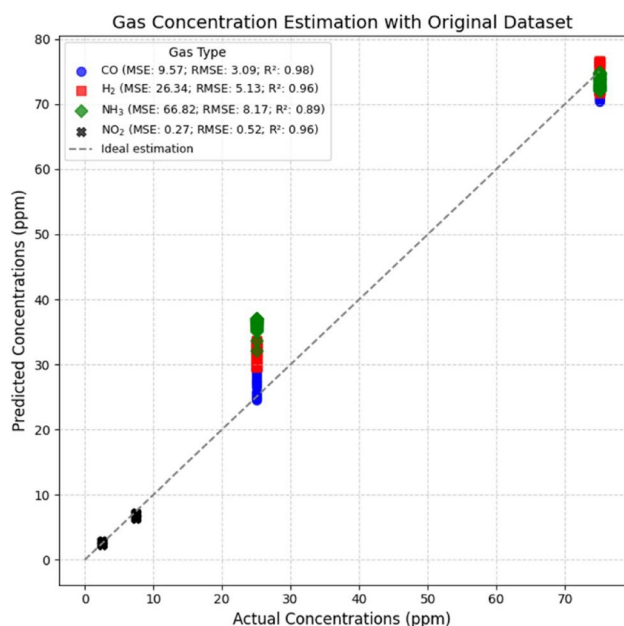


Fig. 8 A scatter plot compares the predicted and actual gas concentrations for the test set using a linear regression model trained on the original dataset. The results are shown for four gases: **a** CO, **b** H₂, **c** NH₃, and **d** NO₂. The dashed line represents the ideal estimation, while different colored markers indicate the predicted values for each gas type. The model performance is evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score for each gas

model to effectively capture its concentration patterns. CO and H₂ also demonstrate relatively high accuracy, with RMSE values of 3.0933 and 5.1322, and R^2 scores of 0.98 and 0.96, respectively. These results imply that the model can adequately handle the sensor signals of these two gases, though some minor discrepancies still exist. In contrast, NH₃ presents the highest RMSE of 8.1746 and the lowest R^2 score of 0.89, reflecting greater difficulty in accurately predicting its concentration. This could be attributed to overlapping sensor responses between NH₃ and other reducing gases or the presence of noise in the sensor signals, which limits the model's ability to differentiate NH₃ effectively. Moreover, while data augmentation was employed to improve the model's generalization, particularly through concentration interpolation, it may have introduced synthetic patterns that unintentionally enhanced class separation between H₂ and NH₃ at higher concentrations. The interpolation was performed using smooth fitted curves derived from low- and mid-range data, rather than strict linear extrapolation, which likely altered the slope behavior observed in the augmented dataset. Although this improved discrimination performance during classification, it may also have distorted the regression relationships, especially where sensor responses were originally similar.

Overall, these results highlight the potential of linear regression in gas concentration estimation, particularly for gases with distinct sensor responses like NO₂. However, the reduced performance for NH₃ emphasizes the need for more advanced strategies, such as sensor fusion, feature engineering, decorrelation techniques (e.g., PCR, PLS), or hybrid modeling approaches, to improve the model's ability to handle complex and overlapping gas signals. Future work could also explore the integration of deep learning models or the incorporation of temporal information to enhance the system's robustness and adaptability in real-world applications, such as environmental monitoring and industrial safety control.

4 Conclusion

This study demonstrates that a minimal MOS sensor array, combined with machine learning and data augmentation, can effectively classify hazardous gases and estimate their concentrations with high accuracy. Among the models, Gradient Boosting (GB) achieved perfect classification, while Decision Tree (DT) and Quadratic Discriminant Analysis (QDA) showed high accuracy with minor misclassification. However, the models faced challenges in distinguishing between reducing gases like NH₃ and H₂. Data augmentation significantly improved the performance of GB and DT, while Random Forest (RF) suffered from overfitting, leading to poor generalization. In gas concentration estimation using linear regression, the model achieved high accuracy for NO₂ with minimal error, while NH₃ exhibited the highest error due to overlapping sensor responses and limited feature extraction capability. These findings highlight the effectiveness of data augmentation in enhancing model robustness and the need for more advanced techniques to handle complex gas mixtures. Future studies could explore deep learning approaches, including CNNs and RNNs, to improve feature extraction and overcome the current challenges in distinguishing between H₂ and NH₃. Additionally, incorporating advanced data augmentation techniques, such as generative adversarial networks (GANs) or physics-based synthetic data modeling, could further improve model generalization and scalability for real-world applications, such as industrial safety monitoring and environmental pollution control.

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Author Contribution Nguyen Ngoc Viet conceived and designed the experiments, performed measurements, processed the data, and wrote

the main manuscript text. Nguyen Viet Chien was responsible for setting up the measurement systems. Le Minh Huy provided support for the implementation and validation of AI models. Nguyen Van Hieu, along with all co-authors, contributed to the manuscript review and provided constructive feedback.

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Data Availability Data is provided within the manuscript.

Declarations

Conflict of Interest The authors declare that they have no conflicts of interest in this work.

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